

SAFETY QUIZ:CHE382-Chemical Engineering Laboratory II

Please answer the following questions testing your knowledge from 6 selected chapters of the JT Baker Hazardous Chemical Safety document; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards and Compressed Gases. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words. (5 points per question)

1. What is the general definition of a toxic chemical?

Any chemical that causes damage to structure or disturbance to function when ingested, inhaled, absorbed, applied to, injected into or developed within the body.

2. Name the general types of damage that a toxic chemical can cause:

A toxic chemical can cause injury by damaging biological structure or disturbing biological function.

3. Name two parameters that can effect toxicity:

Mode of entry and dose.

4. What is the DOT (Department of Transportation) definition of a nonflammable compressed gas?

Nonflammable materials which have a psia greater than 40 at 70F or psia greater than 104 at 130F.

5. a) How does one determine how much material is remaining in a non-liquified gas cylinder?

By taking the cylinder pressure gauge reading as the amount is directly proportional to the pressure inside the cylinder.

- b) On page 13-10, which pressure gauge on the two-stage regulator will give you the cylinder pressure reading?

The gauge on the right.

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6. Other than a single compressed gas, what else can be stored inside a compressed gas cylinder?



Liquified gas and non-liquified gas

7. Why do insidious hazards represent a problem to workers?

Because they are not usually seen, smelled, tasted or felt. Workers may never be aware that a problem exists until a chronic, systemic poisoning has occurred.

8. What are the four general categories of insidious hazards?

Unanticipated chemical reactions, unanticipated toxicity, improper facility design, improper handling of equipment or chemicals.

9. Name 2 control measures that can be implemented to reduce insidious hazards in the laboratory.

Include an insidious hazard section in safety audit, and thoroughly train all operators in the hazards of substances commonly used in the lab.

10. What is the definition of a reactive chemical?

Substances which enter into violent reactions during which the spontaneous liberation of heat and gasses is too rapid to be safely dissipated by the surroundings, which creates a reaction that is difficult to control.

11. Name the two groups and their sub-headings used to categorize reactive chemicals:

The first group is unstable compounds which include explosives such as TNT and others such as monomers.

The second group is unstable mixtures which include explosives, water-reactive compounds, air-sensitive compounds, and other mixtures of oxidizing agents with reducing agents.

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12. List two precautionary measures for handling and/or storing reactive chemicals.

Have plenty of water available for flushing where water-sensitive chemicals are not involved, and protect reactive chemicals, especially explosives, from shock.

13. How are corrosive chemicals generally defined?

Substances that, by direct chemical action, are injurious to the body tissue or corrosive to metal.

14. How do corrosive chemicals act on body tissues?

Through direct chemical reaction, dissolution of essential components, destruction or protein and disruption of cell membranes.

15. True or False: Concentrated alkaline solutions have a more corrosive effect on tissue than most strong inorganic acids.

True

16. Name the six categories of common corrosive liquids.

Inorganic acids, organic acids, other inorganics, caustic solutions, organic solvents, other organics.

17. What category of corrosive solids presents the greatest hazard because of their frequent use in industry?

Caustic alkalies

18. Name the three components that make up the FIRE TRIANGLE:

Fuel, oxidizer, and ignition source.

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19. What category of flammables is the most common source of fires in industry?

Flammable solvents.

20. List two common safety problems and two safety control measures for flammable solvents.

Improperly store glass containers of flammable solvents and mixed solvents in a waste disposal are two common safety problems.

Isolating solvents from reactive chemicals and transferring solvents in a working fume hood are two common control measures for flammable solvents.

BONUS QUESTION:

Where is the eyeglass cleaning station located in the Unit Operations Laboratory (Room 120)?

On the wall connecting to the classroom next door, by the process control lab equipment.

Comment Summary

Page 1

1. -2
local, acute, chronic and systemic toxicity
2. it's the one on the left
-2.5

Page 2

3. -2
gas dissolved in liquid or solid, and compressed gas mixture